

Use the scenario below for questions 1 – 5.

Tom purchased a used Honda CRV from a dealership for \$15,575. The car depreciates 15% per year. The situation can be modeled by the function $P(x) = \$15,575(0.85)^x$ where x represents the number of years since Tom purchased the car and P is the value of the car.

1. Does the equation $P(x) = \$15,575(0.85)^x$ represent exponential growth or decay?

decay since $b < 1$

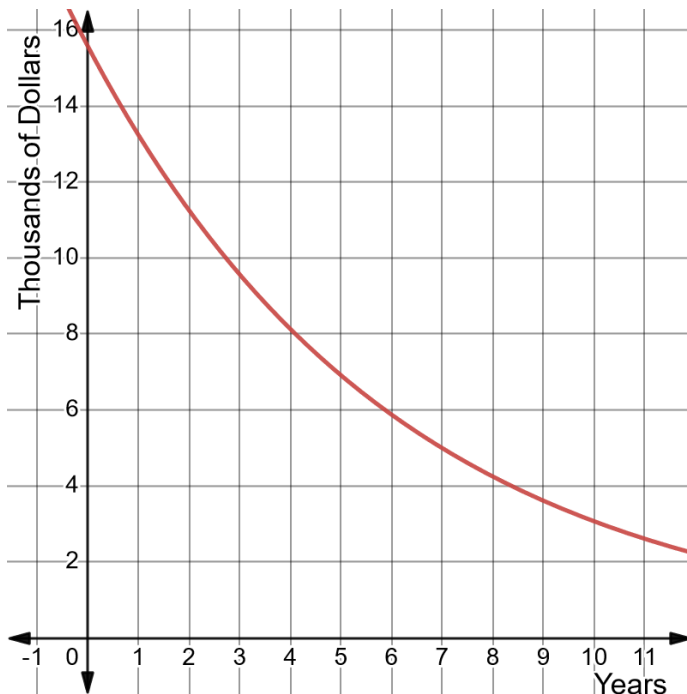
2. What is the y -intercept and what does it represent in this situation?

It represents the initial amount paid for the car.

3. Explain why the value of b is not 0.15, the rate of depreciation.

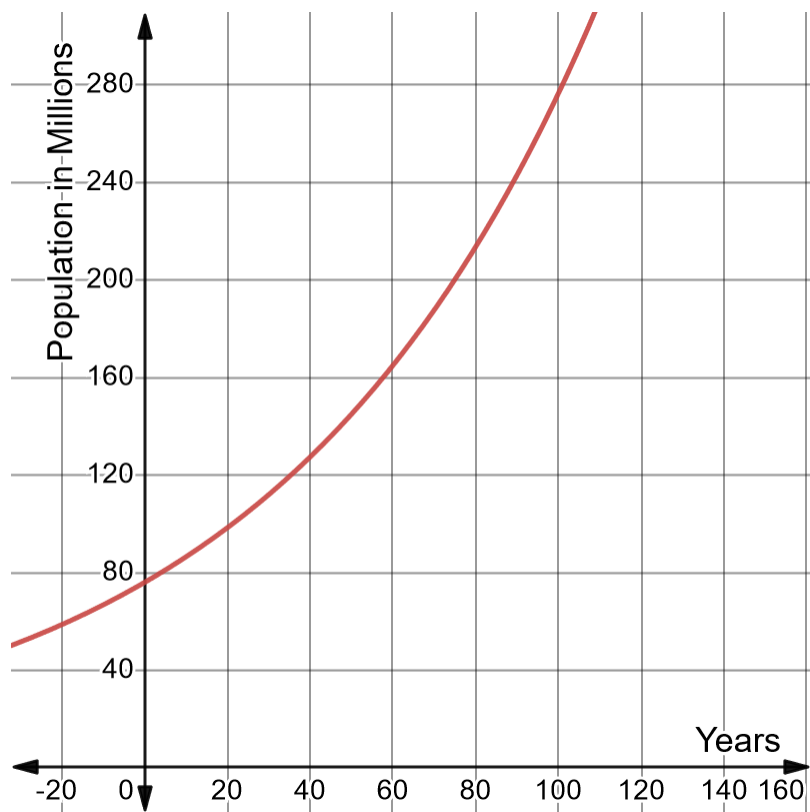
Because $1 - 0.15 = 0.85$ is what the car is still worth.

4. Graph the equation $P(x) = \$15,575(0.85)^x$.



The equation $f(t) = 76(1 + 0.013)^t$ represents the U.S. population and t represents the number of years after 1900. Graph the function below, then determine the key features and interpret how each relates to the context.

5.



Key Features		Interpretations
6. Domain	$x \geq 0$	The years after 1900
7. Range	$y \geq 76$	The population of the U.S.
8. Y-intercept	(0,76)	In 1900, the population was 76 million
9. Constant Percent Rate of Change	$1.013 = 101.3\%$	The population increased yearly by a rate of 101.3%.
10. End Behavior	As $x \rightarrow -\infty, y \rightarrow 0$ As $x \rightarrow \infty, y \rightarrow \infty$	The population can continue to grow exponentially over the years.